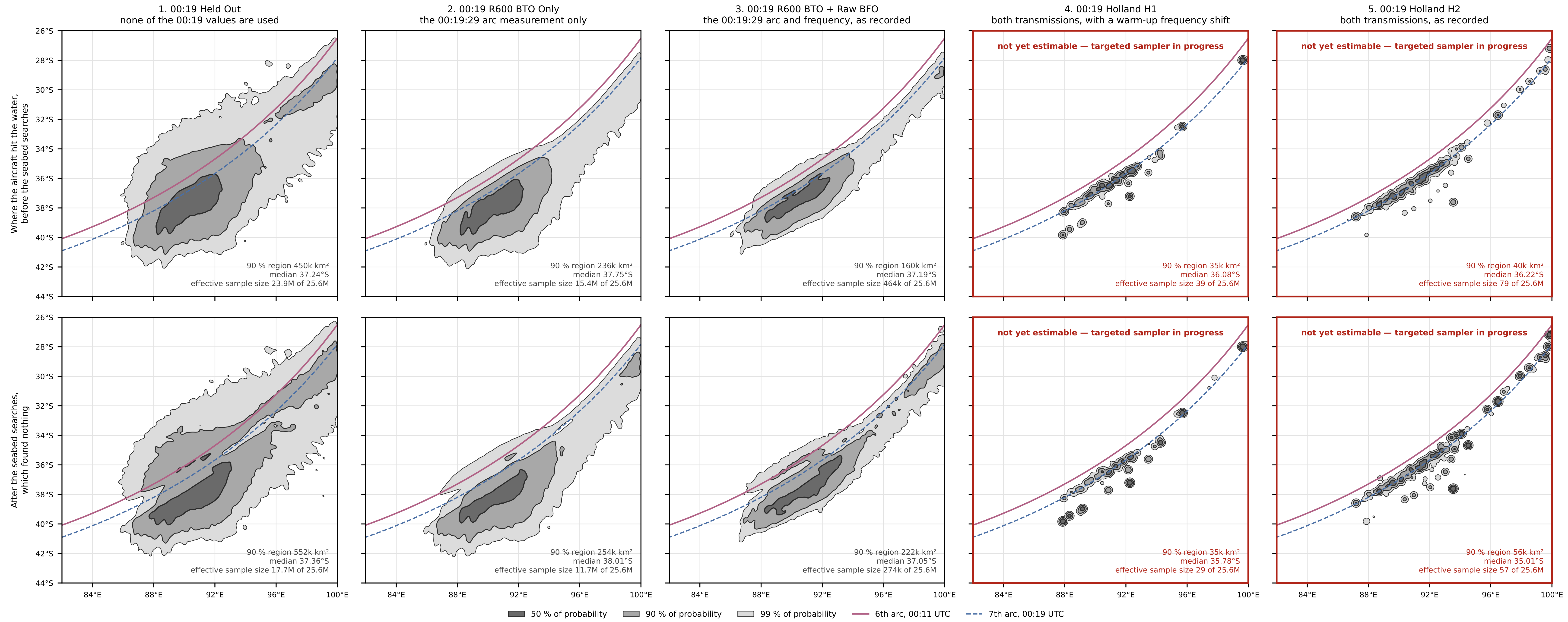
1. 00:19 Held Out
none of the 00:19 values are used
2. 00:19 R600 BTO Only
the 00:19:29 arc measurement only
3. 00:19 R600 BTO + Raw BFO
the 00:19:29 arc and frequency, as recorded
4. 00:19 Holland H1
both transmissions, with a warm-up frequency shift
5. 00:19 Holland H2
both transmissions, as recorded
Where the aircraft hit the water, before the seabed searches
After the seabed searches, which found nothing
26°S
28°S
30°S
32°S
34°S
36°S
38°S
40°S
42°S
44°S
84°E
88°E
92°E
96°E
100°E
not yet estimable — targeted sampler in progress
90 % region 450k km²
median 37.24°S
effective sample size 23.9M of 25.6M
90 % region 236k km²
median 37.75°S
effective sample size 15.4M of 25.6M
90 % region 160k km²
median 37.19°S
effective sample size 464k of 25.6M
90 % region 35k km²
median 36.08°S
effective sample size 39 of 25.6M
90 % region 40k km²
median 36.22°S
effective sample size 79 of 25.6M
90 % region 552k km²
median 37.36°S
effective sample size 17.7M of 25.6M
90 % region 254k km²
median 38.01°S
effective sample size 11.7M of 25.6M
90 % region 222k km²
median 37.05°S
effective sample size 274k of 25.6M
90 % region 35k km²
median 35.78°S
effective sample size 29 of 25.6M
90 % region 56k km²
median 35.01°S
effective sample size 57 of 25.6M
50 % of probability
90 % of probability
99 % of probability
6th arc, 00:11 UTC
7th arc, 00:19 UTC

WHAT THIS SHOWS. Each column makes a different choice about the two radio transmissions from the aircraft at 00:19 UTC. The top row shows where the aircraft hit the water before the seabed searches. The bottom row shows the same after the searches, which found nothing. Dark to light, the three bands hold 50, 90 and 99 per cent of the probability. The searches usually make the area LARGER, not smaller: they take away a block in the middle of the corridor and leave a ring around it. A search that finds nothing tells you where the aircraft is not; it does not tell you where it is. Columns 4 and 5 cannot be calculated correctly from this set of flight paths yet, and a better method is being prepared, so their shapes and numbers are not results. Every number on this page comes from a chain of models that is not yet stable, so treat all of it as provisional.

TECHNICAL. Observations (engine/data/satcom-observations.csv, released unredacted SITA/Inmarsat logs): 00:19:29.416 UTC R600 log-on request, BTO 18,400 us (raw 23,000 less the standard 4,600 us log-on-channel correction, sd 63 us) and BFO 182 Hz (tabulated sd 7 Hz); 00:19:37.443 UTC R1200 log-on acknowledge, BFO -2 Hz. The R1200 BTO is excluded as anomalous. The constant BFO bias is Davey's 150 +/- 25 Hz prior, marginalised per particle, so the two 00:19 BFOs are scored jointly. Internal arms, in column order: none, r600-bto, r600_no-offset, both_startup-offset x fuel-exhaustion, both_no-offset x other, all with end of flight's +alive existence constraint. Holland's start-up offset is arXiv:1702.02432v3 section V: 17-130 Hz on the acknowledge and a further 0-6 Hz on the request. Seabed-search likelihood: ATSB Phase 2 union 120,486.5 km2 plus Bluefin-21/Artemis 771.4 km2, coverage rasterised at 0.01 deg, detection probability 0.945 and 0.900 conditional on a detectable target, undetectable fraction rho = 0.05, point target, shared miss dependence; Ocean Infinity layers excluded. PRIOR c-next-c-repro-radar: 289.7° initial track at 2014-03-07T18:01:49Z, 8 seed files, 25,600,064 impacts in all. Bands are 50/90/99 % highest-posterior-density regions on a 0.02 deg grid smoothed at 0.1 deg (6 NM); areas on the authalic sphere. ESS is the Kish effective sample size of the importance weights; below 1,000 the region is not estimable and is reported as such. LABELS run C NOT CONVERGED (free split-half 0.918 vs 0.924); core run C P(family) fixed (free 0.589, Davey dynamics + radar 0.252, descent-climb 0.135, routes 0.024); end of flight: rapid descents above 6,500 ft/min and unloading not reachable by the model, residual roll P(left) 0.8 PROVISIONAL, trim at loss and two-tank takeover not applied; Holland H1/H2 not estimable; compact impacts rebuilt for evaluate by the Pleiades stub; point target only (no run C wreckage samples); all rows incl. contact speed above 1.25 VMO; architecture stand-in run